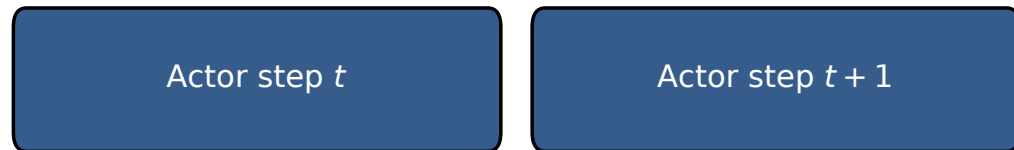


(a) Benefit mechanism

Larger benefit when hit rate p is high
and the speculator is faster ($\alpha \gg \beta$).

Sequential



prefetched result
ready in cache

$$\text{reduction} \approx \frac{p}{1+p} \times \frac{\alpha}{\alpha+\beta}$$

Speculative

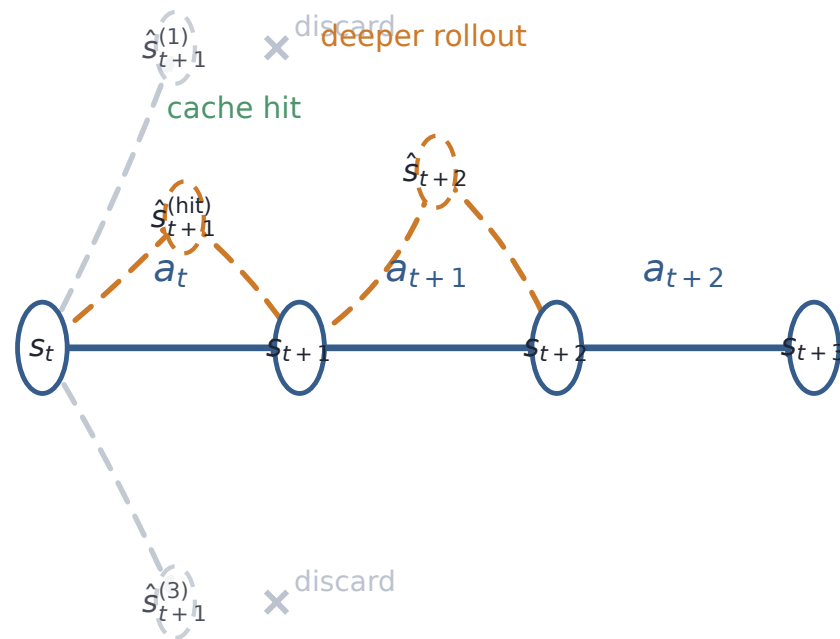


hidden wait /
time saved



time

(b) Agent trajectory



Illustrative schematic derived from the dialogue summary; not to scale.